

Tutorial - 2

Topics: Matrices, Simultaneous Linear Equations

N.B. The first part of this tutorial we will revise matrix multiplication and related concepts.

1. Let A be a 3×3 matrix such that $A = -A^\top$. Consider the homogenous system of linear equations $Ax = 0$. Prove that the system has non-trivial solutions.
2. Let A and B be $n \times n$ matrices with $AB = 0$. Prove the following:
 - (a) $\text{rank}(A) + \text{rank}(B) \leq n$
 - (b) If $A \neq 0$ and $B \neq 0$ then, if the non-homogenous system $Ax = b$ is consistent (similarly, if $Bx = b$ consistent) for some $b \in \Re^3$, then it has infinite solutions.
3. Let A be an $n \times n$ matrix all of whose entries are integers. Prove that there is a matrix B with integer entries such that $AB = BA = I$ if and only if $|A| = 1$ or $|A| = -1$
4. Let $n \geq 2$ and let A be a non-singular $n \times n$ matrix. Prove that: $(\text{adj} A)^{-1} = \text{adj}(A^{-1}) = \frac{A}{|A|}$
5. Let A be a 4×3 matrix, and let b and c be two vectors in $\Re^4$. Suppose the non-homogeneous system $Ax = b$ has a unique solution. What can you say about the number of solutions of the system $Ax = c$
6. Consider an $n \times m$ matrix A of rank n . Show that there exists an $m \times n$ matrix X such that $AX = I_n$. If $n < m$, how many such matrices X are there?